Low power sensor expansion

Options for the Feather ESP32 V2 and SparkFun AS3935 station
Planning overview | Reviewed 11 October 2026 | Proposed additions, not implemented

Yes, useful additions are possible without opening the box to the weather. Start with internal temperature/humidity for enclosure health, then a battery gauge if charge estimates matter. Pressure is another promising measurement. Ambient weather probes need a separate sheltered sensing head.

Choose the measurement first

Candidate	Useful reading	Seal and power implications
SHT40 Temperature / humidity [1]	Internal heat and humidity trends; useful for enclosure monitoring, but cannot certify dryness or detect every leak.	No new hole. Keep away from heat sources and silica. Single measurements; switched supply or low-current idle.
MAX17048 Battery gauge [3,7]	Battery voltage and model-based charge estimate; better information than a generic voltage lookup, still an estimate.	No new hole. Battery-connected gauge plus I2C. Check breakout LED, pull-ups and power domains.
BMP390 Pressure [2]	Station pressure and trends; sea-level pressure needs altitude correction. Not a storm warning.	Can stay inside if the vent equalizes pressure adequately. Forced single measurement, then sleep or power off.
External probe Temperature / humidity	Actual ambient air readings, rather than conditions inside the enclosure.	Radiation shield, airflow, sealed gland, strain relief and drip loop. Short cable or suitable external interface.

The vent does not make a weather screen

The hydrophobic pressure vent passes air [5], making internal barometric sensing plausible. Validate equalization against a reference; a wet or blocked vent can delay it. A completely airtight case can instead show pressure changes caused by its own temperature.

Internal humidity reflects trapped air, desiccant, heating and slow exchange. Label it **inside enclosure**. A dry reading near silica can hide dampness elsewhere. For ambient measurements, the sensor needs representative airflow and separation from heat sources [6]. Internal dew point cannot rule out condensation on colder surfaces.

Other possibilities

An internal leak contact can detect water at the case floor; intermittent sensing limits drain and corrosion. Light/UV needs a sealed window or external head. Wind/rain contacts need outdoor hardware and reliable sleep-time counting. Heated gas sensors, fans, continuous positioning and lit displays need a separate power budget.

Power and electrical design

Keep lightning listening continuously; sample slow sensors briefly and batch their uploads.

Two different power policies

Keep the AS3935 on the Feather's **main 3V supply** with its existing SPI wiring, CS 27 and interrupt 33. Added I2C sensors can use the V2 **SDA/SCL labels or board constants**; do not copy pin numbers from another Feather model.

The switched QT supply is controlled by **NEOPIXEL_I2C_POWER** [4]. Current firmware holds it LOW, so a plugged-in QT sensor will not work without firmware changes. A future driver enables it, waits for startup, configures and reads devices, then powers down before sleep. It also powers the NeoPixel; keep that dark.

A MAX17048 gauge should stay battery-powered between reads. On the Adafruit board, its two JST ports connect battery and load/charger in parallel; the chip is powered by the battery, not QT/VIN. It still needs I2C and correct battery polarity. Its green LED has a disable jumper [7]. Verify the exact board before wiring.

Review the shared bus power domain. Pull-ups from an always-powered device can feed an unpowered sensor; BMP390 explicitly prohibits high interface levels with its interface supply off [2]. Use compatible power domains or isolation, release bus pins before shutdown and measure leakage. Check addresses, parallel pull-ups and bounded recovery from a stuck bus.

The upload schedule often matters more

Start with existing six-hour check-ins and already-required upload wakes. Measure before Wi-Fi heating and add telemetry to that upload. This suits slow enclosure trends, not prompt leak alarms. Preserve lightning capture priority and avoid reading every add-on on noise interrupts.

Optional 15-60 minute logging could wake, measure and store with Wi-Fi off, then upload in a batch. Each boot still costs energy and needs firmware support. Uploading every 15 minutes instead would increase quiet sessions from **4 to 96 per day**, not necessarily total energy by the same factor.

Chip figures are not whole-board budgets

Device	Typical chip figures
SHT40, heater off [1]	About 0.08 uA idle at 25 C; 320 uA during measurement. Keep heaters off for routine readings.
BMP390 [2]	About 1.4 uA sleep at 1.8 V / 1.5 uA at 3.6 V. Forced mode takes one reading then returns to sleep.
MAX17048 [3]	About 3-4 uA hibernate, depending on reset-comparator setting; 23 uA active. Do not assume constant hibernate.

Illustrative battery-side costs: an extra 10 uA = 0.24 mAh/day; 0.2 mA = 4.8 mAh/day. A hypothetical 150 mA, 5-second upload uses 0.208 mAh; 92 extra sessions add about 19.2 mAh/day. These are arithmetic examples, not station measurements. Include LEDs, regulators, boot, conversions and leakage; do not equate a chip's 3.3 V current with LiPo current.

Implementation and acceptance

Development sequence and checks before deciding on a new battery-life estimate.

Preserve the enclosure

Keep the pressure vent unobstructed. Do not use its hole for a cable, trap wires under the gasket or rely on glue alone for a seal. External probes need an appropriate gland, smooth sealing face, strain relief and drip loop. A rated gland does not certify the modified box waterproof. Test the empty case for leaks and dry it fully before reinstalling electronics.

Keep cables short and clear of the ferrite. Long outdoor conductors need surge/ESD and interference review; long I2C is not automatically reliable. An external humidity sensor needs airflow and a radiation shield, not a sealed sensing head. Do not adjust the outdoor instrument during thunderstorms.

Changes needed across the data path

1. Firmware: optional drivers, bounded timeouts and deliberate sampling. Capture pending lightning first; preserve interrupt wake and the low-voltage policy. Failed optional sensors must not stop lightning operation.

2. Records and storage: units, sensor identity, measurement time, fault/validity flags and inside/ambient labels. Keep missing readings distinct from zero. Give queue-format changes a migration; retain sequence identity. Frequent telemetry must not crowd lightning out of the current 128-record queue. Do not use uploadfs as an ordinary upgrade.

3. Cloud and dashboard: extend signed validation, D1 schema, reads, exports and charts together. Batch measurements and preserve honest offline timestamps. Display each reading's age separately from Last station report; distinguish internal from ambient readings and label the charge estimate by source. No new notifications are included.

4. Rollout: update the backend compatibly before flashing; retain firmware rollback. Check mounting space and the exact breakout circuit. Current firmware does not implement these additions.

Acceptance checks

Measure power: compare complete-device sleep current before/after with USB disconnected. Integrate charge for sensor-only wakes, successful uploads and failed retries. Inspect every breakout LED and test bus power leakage.

Keep detection intact: prove lightning interrupt wake and reporting during conversions, absent-device faults, stuck I2C and brownouts. Recheck oscillator tuning and RF noise with add-ons on/off in the final case.

Validate measurements and fit: compare references at the final site, including pressure equalization and altitude correction. Check clearances, battery protection, gasket/vent access and cable restraint. Repeat leak checks and field commissioning in safe weather.

Suggested order: internal SHT40, optional MAX17048, pressure, then external weather. Measure each step. These additions do not make it a certified lightning warning system.

References and detailed plan

[1] Sensirion SHT4x datasheet | [2] Bosch BMP390 datasheet | [3] Analog Devices MAX17048/MAX17049 datasheet | [4] Adafruit Feather ESP32 V2 pinouts | [5] Amphenol VENT-PS1YBK-N8001 | [6] Sensirion humidity/temperature design guide | [7] Adafruit MAX17048 board pinouts

Detailed expansion plan - wiring considerations, energy formula, migration checklist and related project guides.